Wind Analysis Design Calculations

Wind Analysis for Manual Sliding Door - 3 Leaf - RSS - CRM Gate 11 / GATE NO. 25

Project Name:	**Supply and Installation of Manual Sliding Doors**				**NTPL Project Number:**	NTPL-72-PM
A	14.07.2022	KM	MS	AK	VT	FOR INFORMATION
REV	DATE	Prepared by	Checked by Checked by Checked by		Validated by	SUBMISSION PURPOSE

WIND ANALYSIS REPORT

DOC No	Company	Zone	Line	Doc type	Discipline	Sequence	Rev	Version	Status

NIHVA TECHNOLOGIES PVT. LTD.	LOCATION - POSCO MANGAON Wind Analysis for Sliding Door	Project No: NTPL-72-PM Mfg. Year: June 2022

Dead Load

Door size = 7.0 m x 5.1 m = 35.70 sqm

Self-weight of door = 50 kg/sqm x 35.70 = 1785 kg = 1785/100 = 17.85 KN (stability force)

Wind Load

Wind loads on structures shall generally be as per IS: 875-1987-Part 3 based onfollowing parameters: -

Description	Applicable	Remarks
Basic wind speed (Vb)	39 m/s	As per IS-875-1987-Part 3
Terrain Category	2	Assumed Obstructions up to height of 1.5m to 10m.
Risk coefficient (k1)	1.0	As per Table 1 of IS-875-1987-Part 3
Terrain height/structure size factor(k2)	1.05	Shall be suitable taken as per Table 2 of IS-875-1987-Part 3 for relevant class of structure.
Class	A/B/C	
Topography factor(k3)	1.0	Assumed as per Cl.5.3.3 of IS-875-Part3

For computation of wind forces the mean probable design life of all structures shall be taken as 50 years.
Temporary structures shall be designed for a mean probable design life of 25 years and boundary walls as per IS: 875-Part 3
Design wind speed = Vz = k1 xk2x k3 x Vb
Design wind pressure = Pz = 0.6xVz2

If Building is closed structure, external pressure coefficient Cpe shall be calculated as per Table IS 875-Part3.
Internal pressure coefficient shall be as per clause 6.2.3.2 IS 875-Part3.

Design wind speed = Vz = 1.0 x 1.05 x 1.0 x 39 = 40.95 m/s

Design wind pressure = Pz = 0.6 x 40.95 x 40.95 = 1.006 kN/sqm

Wind load = 1.006 x 35.70 =35.91 kN (Sliding Force)

Load Combinations

Listed below are load combinations for analysis & design of Buildings / Structures.
Basic Load Cases

- DL: Dead Load
- WL: Wind Load

Check for Deflection (Serviceability criteria)

Following Load combinations with un-factored loads are to be used for checking deflection for both RCC & Steel Structures & calculation of safe bearing pressure.
Load combinations to check for deflection and calculation of bearing pressure as per IS-875-1987-Part 5 & IS-456-2000

Load Combinations	**DL**	**LL**	**WL**	**EQ**
DL + LL	1.0	1.0	-	-
DL + WL	1.0	-	1.0	
DL + LL + WL	1.0	0.8	0.8	-
DL + EQ	1.0	-	-	1.0
DL + LL + EQ	1.0	0.8	-	0.8

Stabilizing Force = 17 kN

Sliding Force = 35.91 kN

Resultant force = 35.91– 17.85 = 18.06 kN

Connection Design:

SS202 bolts 4 nos of M16 bolt are used for fixing the sliding door at top.

Tension Capacity of bolt = 56.5 kN

Shear capacity of bolt = 31.4 kN

Resultant sliding force = 18.06 kN < 56.50 kN…. Hence connection design is safe in tension.

Stabilizing force = 17.85 kN < 31.40 kN…. Hence connection design is safe in shear.

Hence Sliding door are designed as per Specific requirements, actual design calculations, Factor of Safety with having a sustainable with suitable capacity of Wind Pressure Resistance of 140 Km/ Hr.